

Customer Shopping Behavior Analysis

A comprehensive analysis of **3,900 transactions** across multiple product categories — uncovering spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

DATA SCIENCE

PYTHON · SQL · POWER BI



Dataset Overview

At a Glance

The dataset contains **3,900 rows** and **18 columns**, covering customer demographics, purchase details, and shopping behavior. Only 37 values are missing in the Review Rating column.

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Promo Code, Frequency, Review Rating, Shipping Type

Exploratory Data Analysis in Python

01

Data Loading & Exploration

Imported the dataset using pandas; used `df.info()` and `.describe()` to inspect structure and summary statistics.

Q2 Missing Data Handling

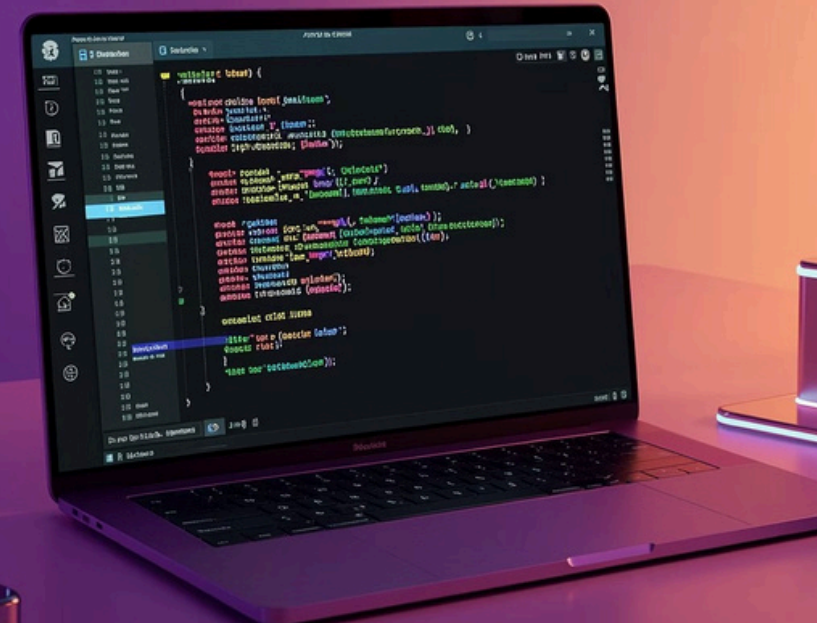
Imputed 37 missing `ReviewRating` values using the **median rating per product category**.

Q3 Standardization & Feature Engineering

Renamed columns to **snake_case**; created `age_group` (binned ages) and `purchase_frequency_days`. Dropped redundant `promo_code_used`

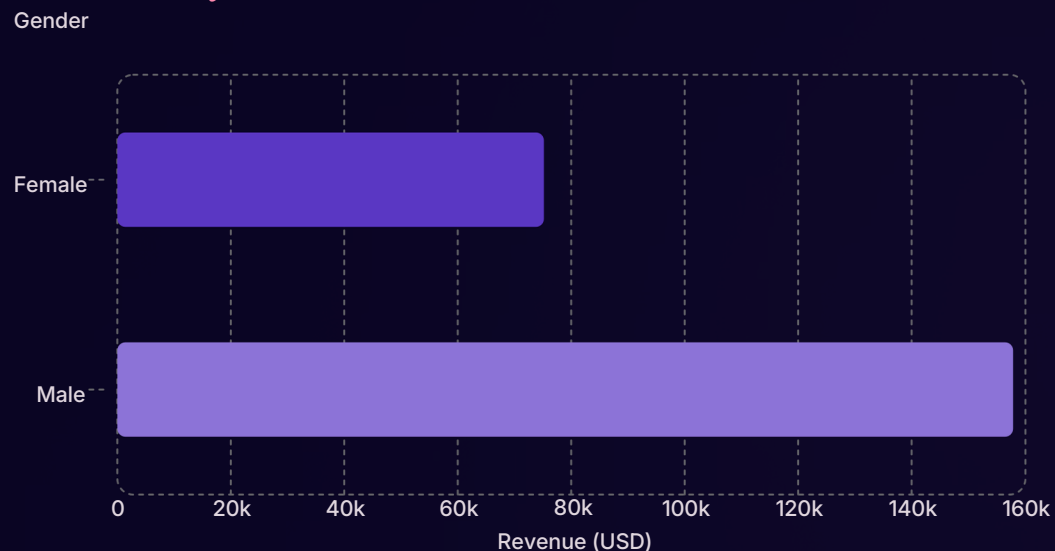
Q4 Database Integration

Connected Python to **PostgreSQL** and loaded the cleaned DataFrame for SQL analysis.



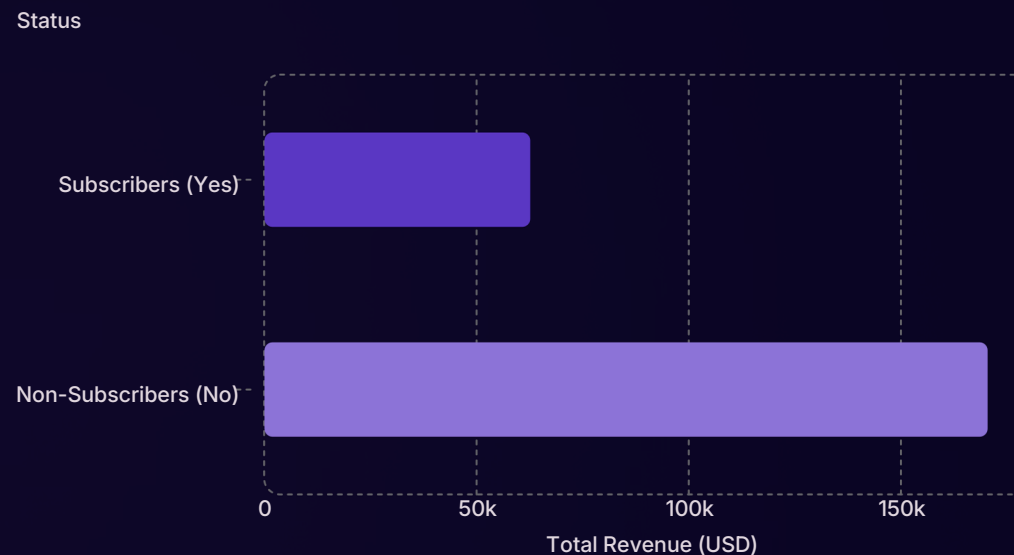
Key SQL Findings: Revenue & Spending

Revenue by Gender



Male customers generated **\$157,890** in revenue — more than double female customers at **\$75,191**.

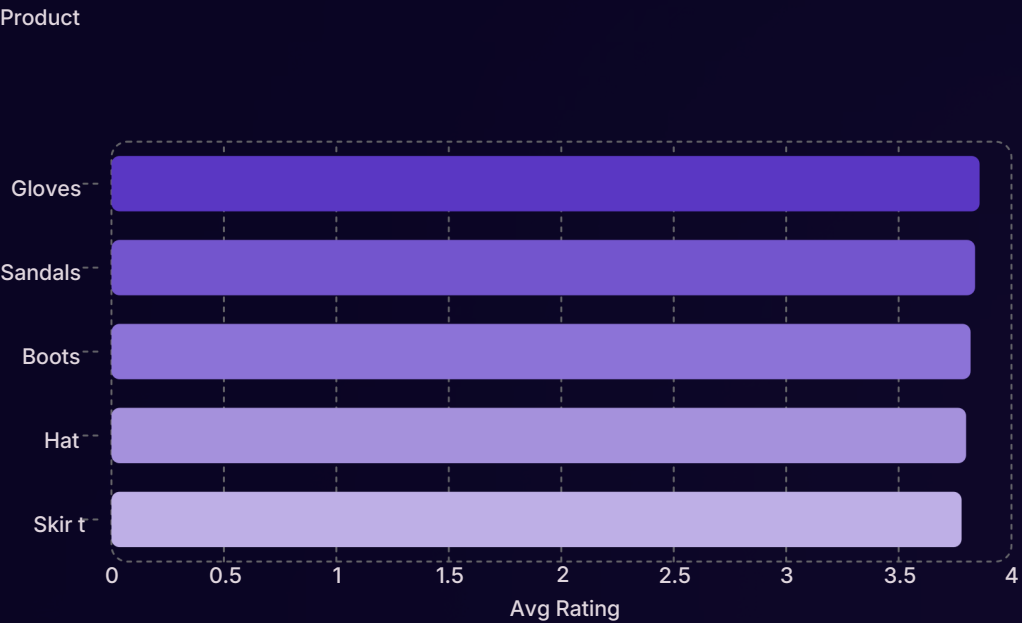
Subscribers vs. Non-Subscribers



Non-subscribers (2,847 customers) drove **\$170,436** in revenue vs. **\$62,645** from 1,053 subscribers — despite nearly identical average spend (~\$59.50).

Product & Discount Insights

Top 5 Products by Average Rating



Products Most Dependent on Discounts

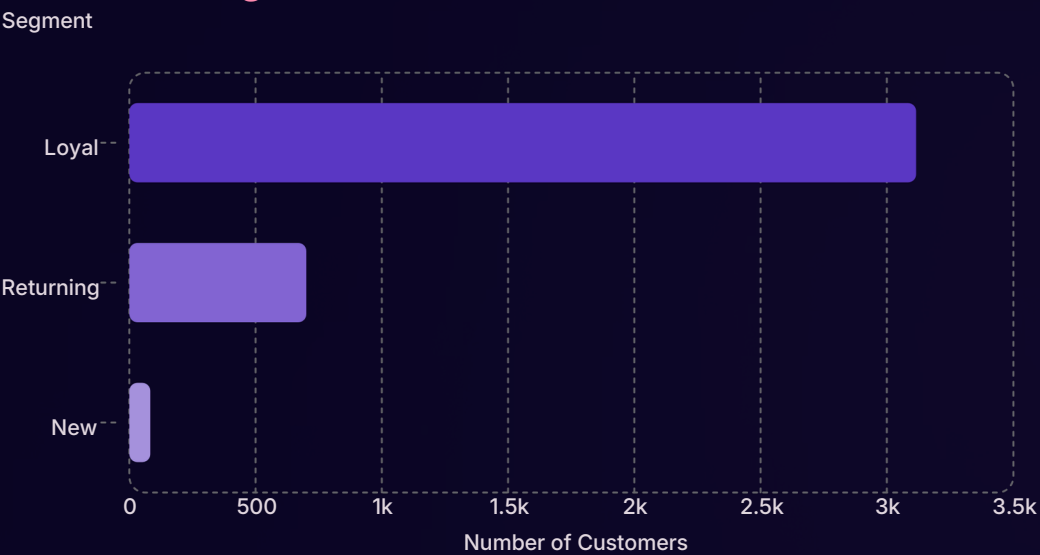


These five products have the highest percentage of discounted purchases — nearly **50%** of sales involve a discount, signaling potential margin risk.

High-Spending Discount Users: 839 customers used discounts yet still spent above the average purchase amount — a valuable segment to retain.

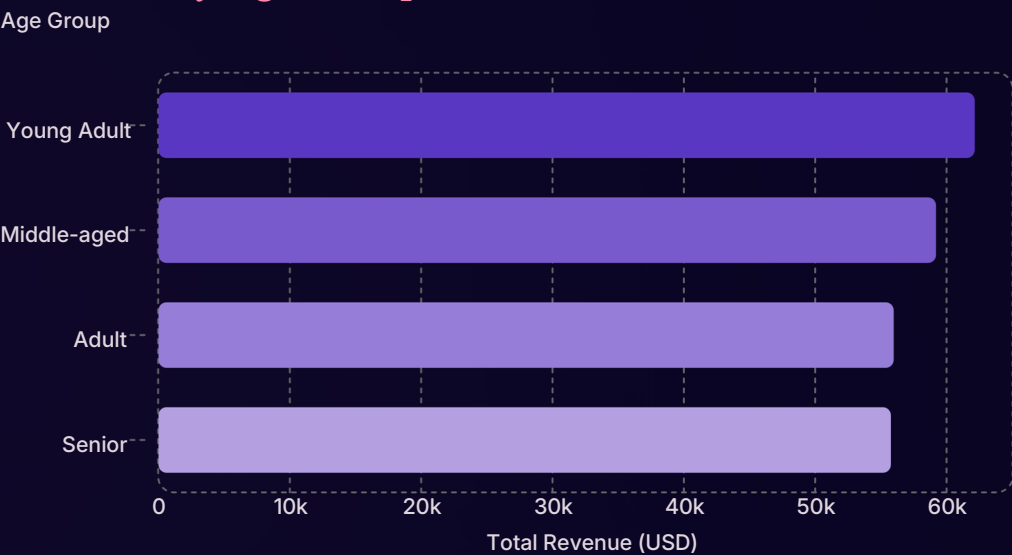
Customer Segmentation & Age Group Revenue

Customer Segments



80% of customers are classified as Loyal. Only 83 are New customers — indicating strong retention but limited new acquisition.

Revenue by Age Group



Young Adults contribute the highest revenue at **\$62,143**. Revenue is relatively evenly distributed across all age groups.

Power BI Dashboard & Business Recommendations

Dashboard KPIs

3.9K customers · **\$59.76** avg purchase · **3.75** avg rating. Filters: Subscription, Gender, Category, Shipping.

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to grow the Loyal segment (currently 3,116 of 3,900).

Review Discount Policy

Balance sales boosts with margin control — ~50% of top products rely on discounts.

Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users.

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in campaigns.